

HW#6

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CHE333

22.1-5 20

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Problem 22.1-2 Gas Solubility in an aqueous Solution

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$$H_{CO_2} = 1.86 \times 10^3 \frac{\text{atm}}{\text{mol}} \quad T = 303 \text{ K} \quad C_{CO_2} = 9.0 \times 10^{-5} \frac{\text{kg CO}_2}{\text{kg H}_2\text{O}}$$

$$P_{CO_2} = H \times C_{CO_2} = (1.86 \times 10^3 \frac{\text{atm}}{\text{mol}}) (9.0 \times 10^{-5} \frac{\text{kg CO}_2}{\text{kg H}_2\text{O}}) \left(\frac{18 \frac{\text{kg H}_2\text{O}}{\text{mol H}_2\text{O}}}{44 \frac{\text{kg CO}_2}{\text{mol CO}_2}} \right) \frac{\text{kg H}_2\text{O}}{\text{kg H}_2\text{O}}$$

$$\rightarrow P_{CO_2} = 0.685 \text{ atm} \rightarrow P_{CO_2} = 6.939 \text{ Pa}$$

Problem 22.1-4

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$$P = 2.02 \times 10^5 \text{ Pa} \quad T = 293 \text{ K}$$

$$P_{SO_2, \text{in}} = 1.52 \times 10^4 \text{ Pa} \quad M_{\text{gas, in}} = 5.70 \text{ kg/mol} \quad M_{\text{H}_2\text{O, in}} = 2.20 \text{ kg/mol}$$

$$L_0 = 2.2 \text{ kmol} \quad V = 5.70 (1 - 0.075) \rightarrow V' = 5.272$$

$$Y_{A_2} = \frac{P_{SO_2}}{P} = \frac{1.52 \times 10^4}{2.02 \times 10^5} = 0.075 \quad A_1 (2.0 \text{ kmol}) = 29.16 \frac{\text{kg}}{\text{mol}} X_{A_1}$$

$$2.2 + 5.272 \left(\frac{0.075}{1 - 0.075} \right) = 2.2 \left(\frac{X_{A_1}}{1 - X_{A_1}} \right) + 5.272 \left(\frac{Y_{A_1}}{1 - Y_{A_1}} \right)$$

$$\rightarrow .428 = 2.2 \left(\frac{X_{A_1}}{1 - X_{A_1}} \right) + 5.272 \left(\frac{14.8 X_{A_1}}{1 - 14.8 X_{A_1}} \right) = \frac{2.2 X_{A_1}}{1 - X_{A_1}} + \frac{78.03 X_{A_1}}{1 - 14.8 X_{A_1}}$$

$$\rightarrow .428 (1 - 14.8 X_{A_1} + 14.8 X_{A_1}^2) = 2.2 X_{A_1} (1 - 14.8 X_{A_1}) + 78.03 X_{A_1} (1 - X_{A_1})$$

$$\rightarrow .428 (1 - 14.8 X_{A_1} + 14.8 X_{A_1}^2) = 2.2 X_{A_1} (1 - 14.8 X_{A_1}) + 78.03 X_{A_1} (1 - X_{A_1})$$

$$\rightarrow .428 - 6.757 X_{A_1} + 6.33 X_{A_1}^2 = -110.59 X_{A_1}^2 + 80.23 X_{A_1}$$

$$\rightarrow 116.92 X_{A_1}^2 - 86.99 X_{A_1} + .428 = 0$$

$$X_{A_1} = \text{quadratic} = 0.005 = X_{A_1} \quad Y_{A_1} = 14.8 X_{A_1} = 14.8 (0.005) \Rightarrow Y_{A_1} = .075$$

$$L_1 = \frac{L'}{1 - X_{A_1}} = \left(\frac{2.2}{1 - 0.005} \right) \quad L_1 = 2.34 \text{ kmol} \quad V_1 = \left(\frac{V'}{1 - Y_{A_1}} \right) = 5.8 \text{ kmol} = V_1$$